



Matheus Paiva Angarola

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EDUCATION

- **University of São Paulo (ICMC - USP)** 2023 - Present
Bachelor's in Computer Science - GPA 3.6/4.0
- **ETEC Martin Luther King (CPS)** Jan 2020 - Dec 2021
Technical Course in System Development - GPA 4.0/4.0

EXPERIENCE

- **Distributed Autonomous Systems Lab (DASLAB) - UIUC** Mar 2026 - Present
Fully Funded Research Intern / Advised by Prof. Girish Chowdhary Champaign, IL, USA
 - Developing adaptive quadruped locomotion policies using a **Mixture of Experts (MoE)** architecture trained via Deep Reinforcement Learning.
 - Investigating **embodied autonomy** by creating scalable models capable of dynamically adjusting to unstructured terrains and physical constraints.
 - Collaborating on state-of-the-art capability-aware navigation systems for field robotics, actively integrating representation learning and visual perception.
- **Mobile Robotics Laboratory (LabRoM)** 2025 - Present
Fully Funded Research Intern / USP Robotics Center (CRob) São Carlos, SP, Brazil
 - Scientific initiation project funded by FAPESP, focused on developing methods for **embodied autonomy** and adaptation of quadruped robot locomotion in diverse scenarios.
- **Robotics Team (SEMEAR Group)** Jan 2025 - Jan 2026
Technical Leader - International Preparation São Carlos, SP, Brazil
 - Led preparation for national and international *RoboCup@Work* competitions.
 - Improved autonomous systems, integrated new sensors, and optimized **navigation, manipulation, and perception algorithms** for dynamic real-world environments.
- **GIVA Laboratory (Applied Image and Computer Vision Group)** Jan 2024 - Dec 2024
ICMC - University of São Paulo São Carlos, SP, Brazil
 - Scientific initiation project funded by FAPESP on machine learning methods for curvature estimation in 3D meshes (relevant for robust 3D/LiDAR representations).
- **Robotics Team (SEMEAR Group)** Jan 2024 - Dec 2024
Project Manager and Programming Supervisor São Carlos, SP, Brazil
 - Managed initiation projects and supervised technical team in *RoboCup@Work*, winning 1st place nationally.
 - Applied advanced robotics concepts, computer vision (YOLO), development with Python, C++, Jetson Nano, and ROS.
- **Robotics Team (SEMEAR Group)** Mar 2023 - Dec 2023
Undergraduate Researcher São Carlos, SP, Brazil
 - Developed an autonomous robot capable of navigating unknown environments; worked on PID control, sensors, and embedded systems.
 - Deepened studies and applied methods in **LiDAR perception, SLAM, and autonomous navigation**.

SKILLS

Programming Languages: Python, C/C++, Java

Robotics & Deployment: Robot Operating System (ROS 1 and 2), Sim2Real Transfer, Edge Computing (NVIDIA Jetson, Raspberry Pi), Hardware Integration, Docker, Linux, Git, AWS

AI & Perception Frameworks: PyTorch, Isaac Lab/Sim, OpenCV, YOLO, Point Cloud Processing

Areas of Interest: Embodied Autonomy, Reinforcement/Imitation Learning, LiDAR/3D Perception, Real-World Robot Deployment, and Field Robotics

Languages: Portuguese (Native), English (Advanced), Spanish (Basic)

PROJECTS

- **Academic Researcher (FAPESP)**

Feb 2025 – Present

- Replaced a single generalist locomotion policy with terrain-specialized expert policies using **Deep Reinforcement Learning**.
- Applied **representation learning** to extract terrain embeddings from onboard signals and clustered them to define terrain families; trained one policy per cluster.
- At deployment, used a nearest-centroid rule to select the appropriate expert online; achieved +16% average improvement over the generalist baseline (higher success rate and stability).
- Accepted for presentation at the *IEEE International Conference on Advanced Robotics (ICAR) 2025*.

- **Academic Researcher (FAPESP)**

Jan 2024 – Dec 2024

- Developed machine learning methods to estimate mean and Gaussian curvature in 3D meshes to reduce processing time.
- Used Python, Scikit-Learn, NumPy, Pandas, and MATLAB for data extraction, modeling, and predictive training.
- Presented results at SIICUSP (USP International Symposium on Scientific Initiation).

PUBLICATIONS

Learning Terrain-Specialized Policies for Adaptive Locomotion in Challenging Environments — Matheus P. Angarola, F. Affonso and M. Becker. *IEEE International Conference on Advanced Robotics (ICAR)*, 2025. **Accepted**. DOI: 10.48550/arXiv.2509.20635.

Towards Capability-Aware Traversability Navigation for Unstructured Environments — G. Capezzuto, Matheus P. Angarola, F. A. G. Tommaselli, M. Becker. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026. **Under Review**.

AWARDS AND RECOGNITIONS

- **RoboCup 2025 @Work** - Salvador, Brazil

2025

- Participated in the *RoboCup 2025 World Championship* in Salvador, Brazil, competing among leading international robotics teams.

- **Brazilian Robotics Competition (CBR) RoboCup @Work**

2024

- First place in the *RoboCup@Work* category of the 2024 Brazilian Robotics Competition.